

Welcome to 2602I Chemistry (Polytechnical Foundation)

Professor Kasper S. Pedersen

Department of Chemistry
Technical University of Denmark

2026

Course overview

Book: *Chemistry 2e* fra OpenStax (Link on DTU Learn announcement)

Exercises: Information will follow on DTU Learn

Course plan: Information will follow on DTU Learn

Lectures: Wednesdays, 10:15-12 in Building 208. auditorium 54+53.

Recordings: Available online.

Teacher: Prof. Kasper S. Pedersen (responsible, kastp@kemi.dtu.dk)

What is chemistry?

Encyclopedia Britannica:

“the science that deals with the properties, composition, and structure of substances (defined as elements and compounds), the transformations they undergo, and the energy that is released or absorbed during these processes”

Why bother?

Some examples where chemistry affects other areas of engineering science:

Construction materials for buildings, bridges,...

Sustainable food

Water quality and pollutants

Medicine

Designed proteins and enzymes

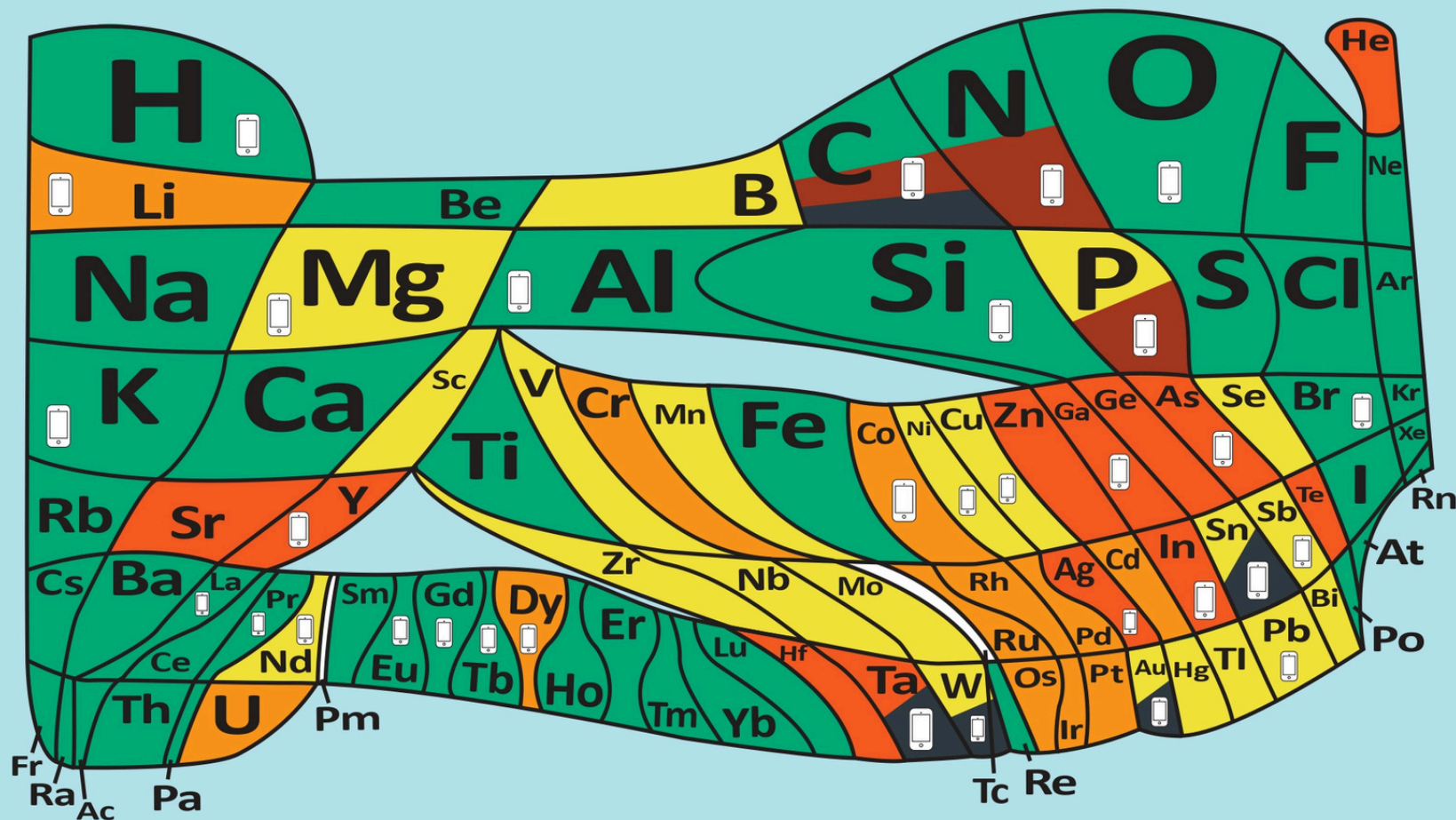
Corrosion

Batteries for your laptop / smart phone / car /...

Climate science

De 90 naturlige grundstoffer, der udgør alt

Hvor meget er der? Er det nok? Er det bæredygtigt?



Tilgængelighed



Bæredygtighed



Grundstoffer brugt i en smartphone

Læs mere og spil videospillet <http://bit.ly/euchems-pt>

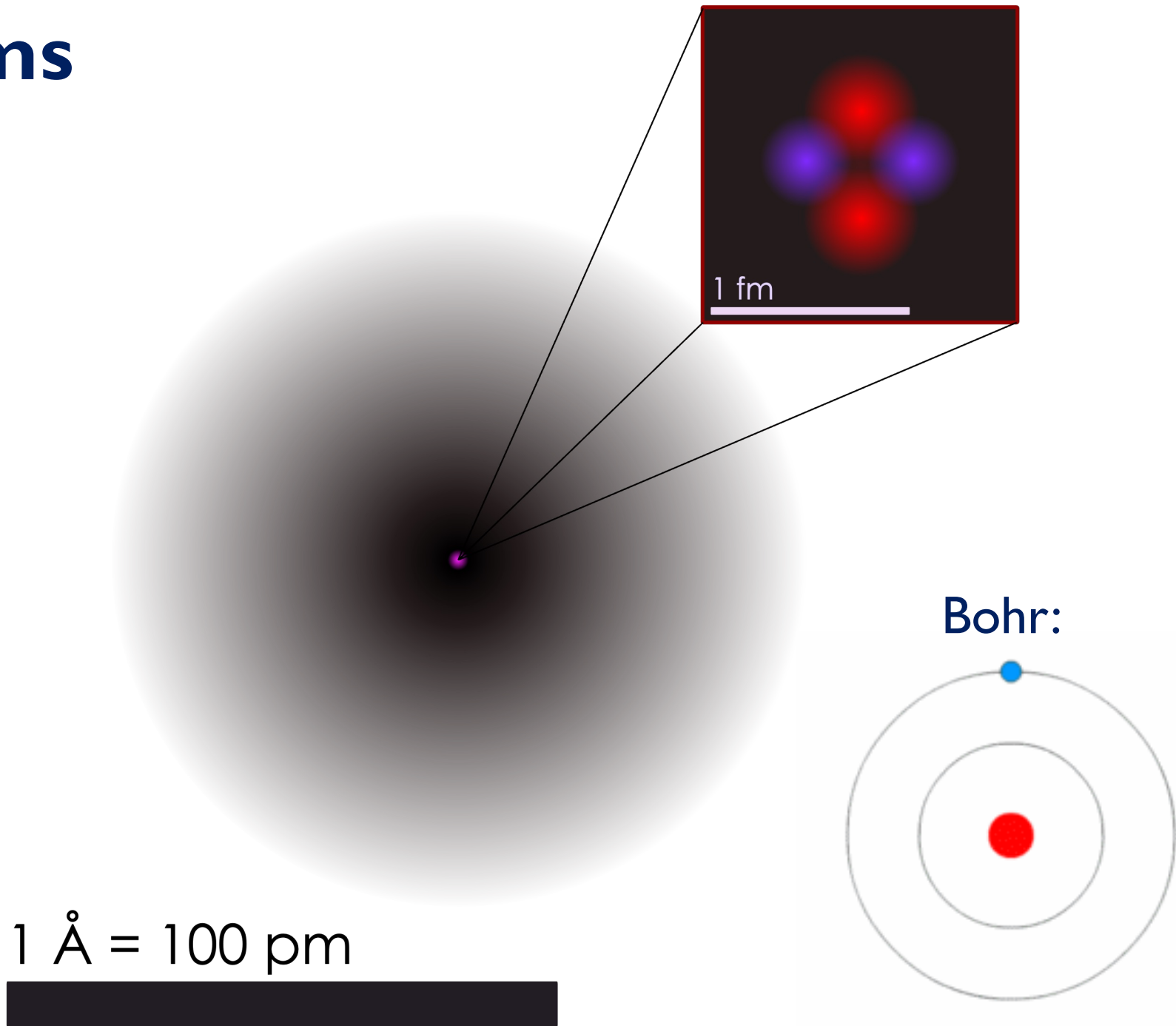
edition 2.0 (2023)



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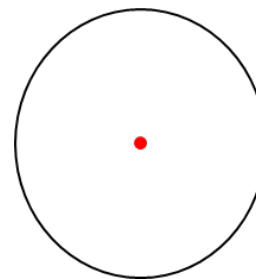
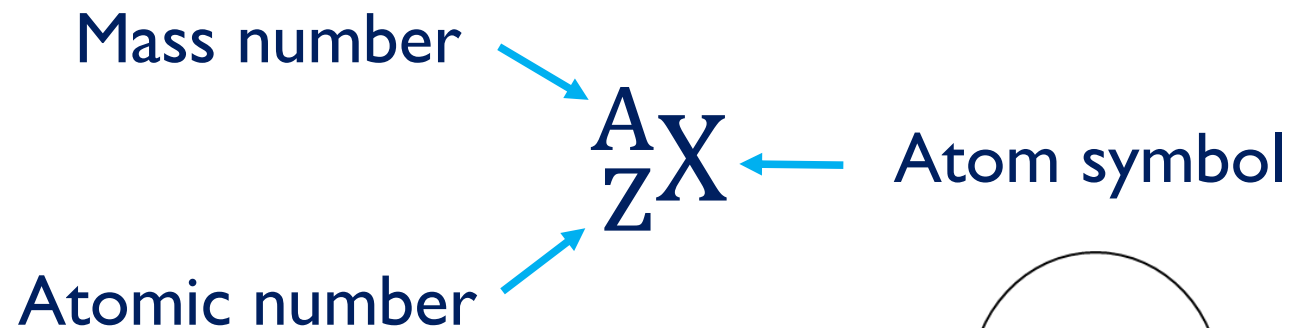
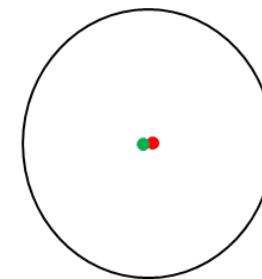
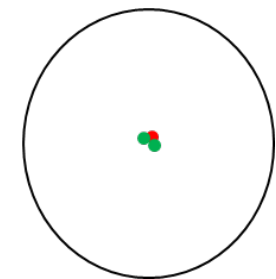
EuChem
European Chemical Society

Atoms



Atoms

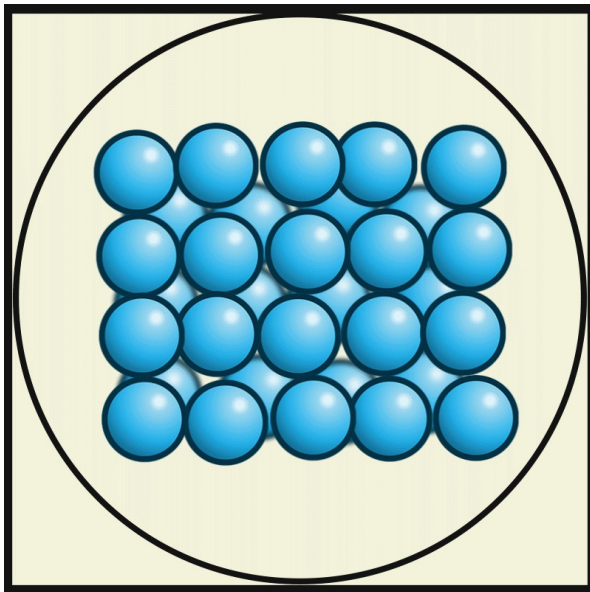
Particle	Mass (g)	Charge (C)	
Electron	9.1×10^{-28}	-1.6×10^{-19}	= “-1”
Proton	1.7×10^{-24}	$+1.6 \times 10^{-19}$	= “+1”
Neutron	1.7×10^{-24}	0	= 0


 ${}^1_1\text{H}$

 ${}^2_1\text{H}$

 ${}^3_1\text{H}$

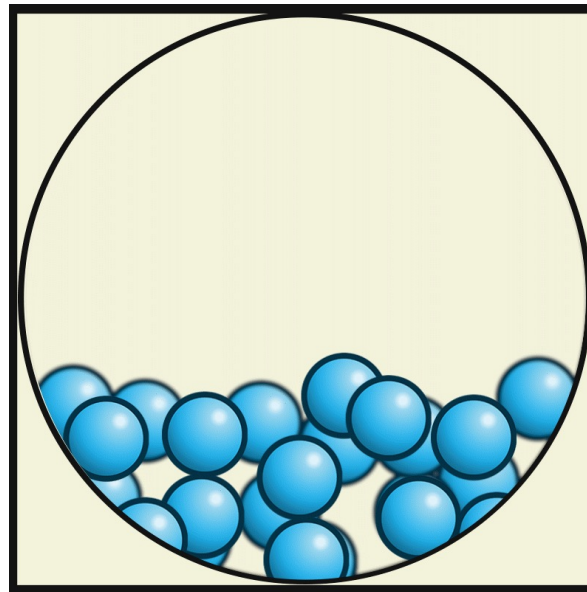
Experiment

- ▶ Sinking ice cubes
- ▶ Sublimation of CO_2

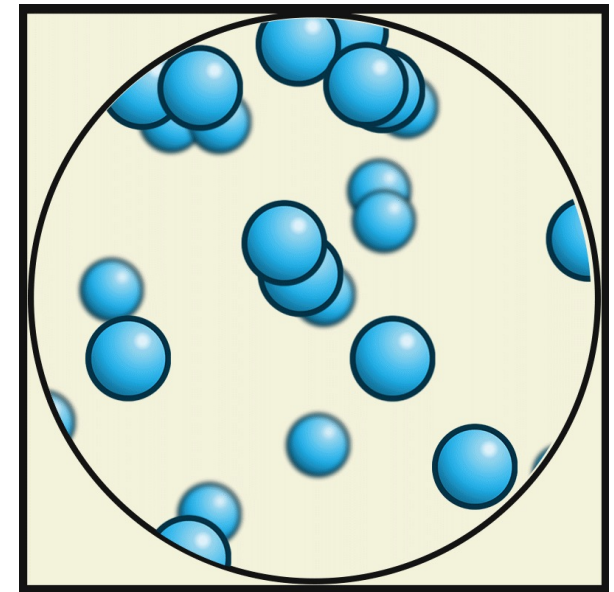
The three states of matter



Solid



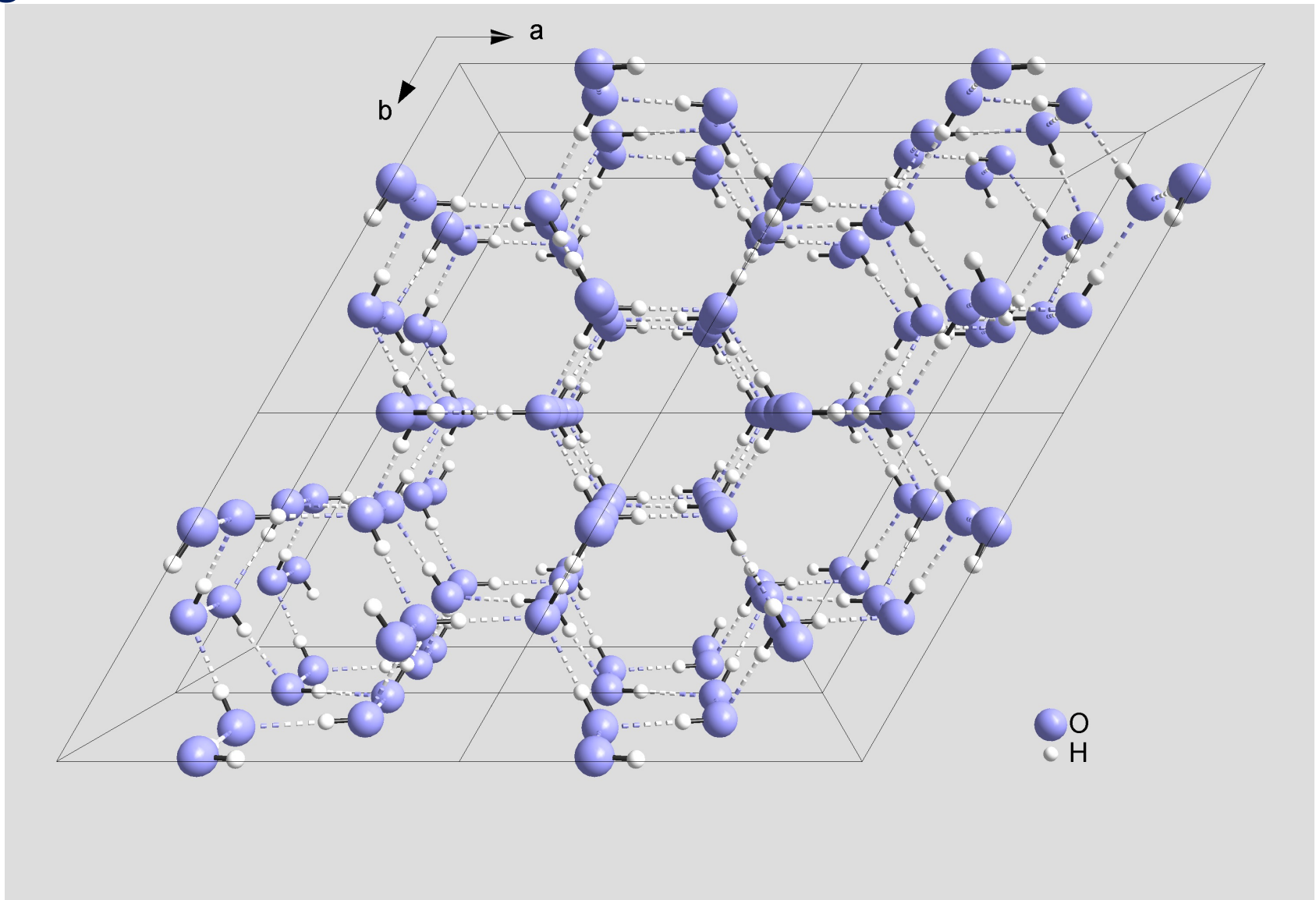
Liquid



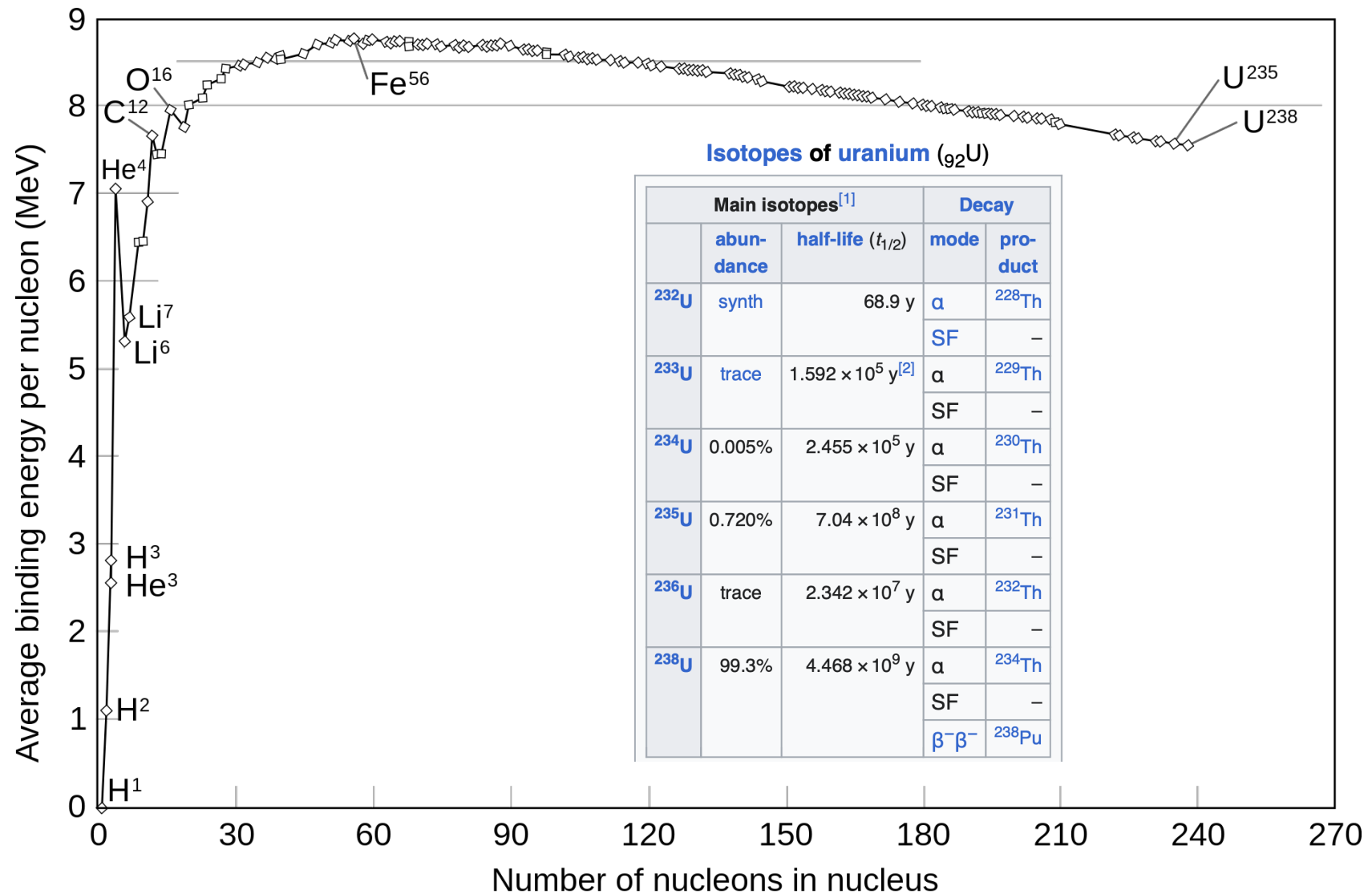
Gas

Experiment

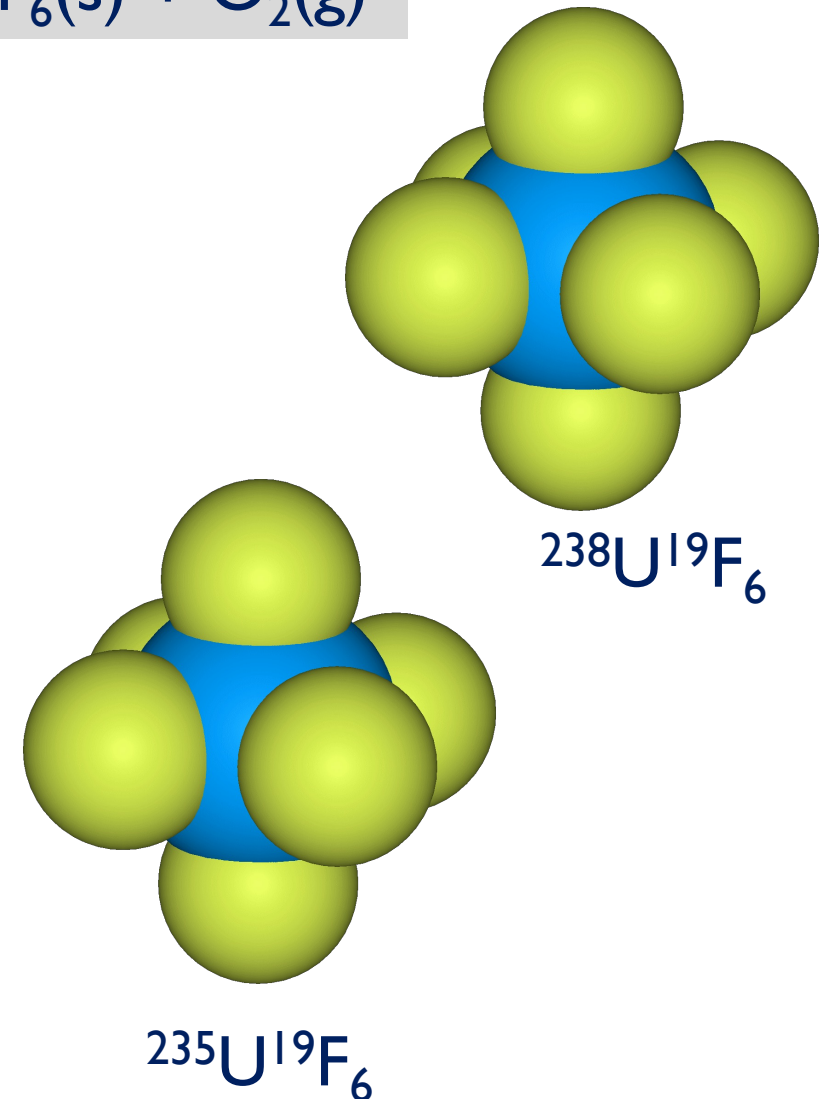
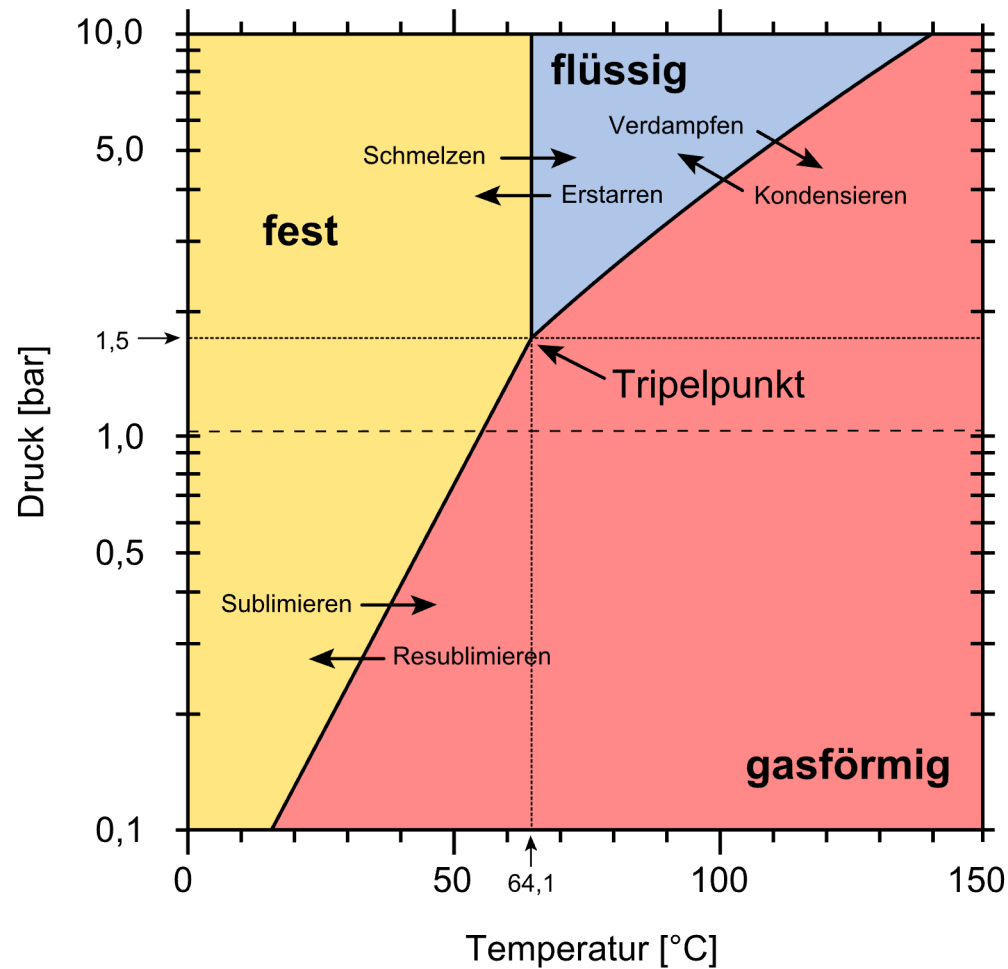
Sinking ice cubes



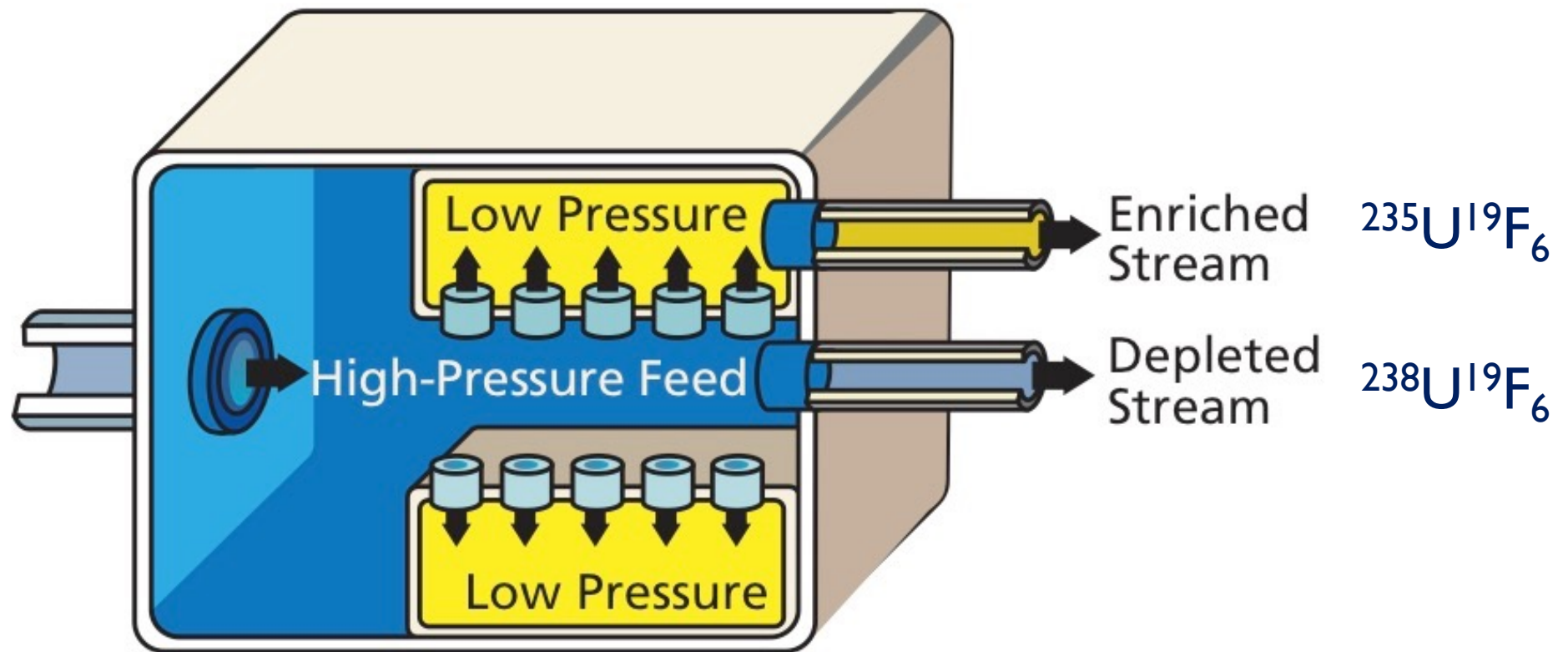
Nuclear energy: fusion and fission



UF₆ and nuclear fuel



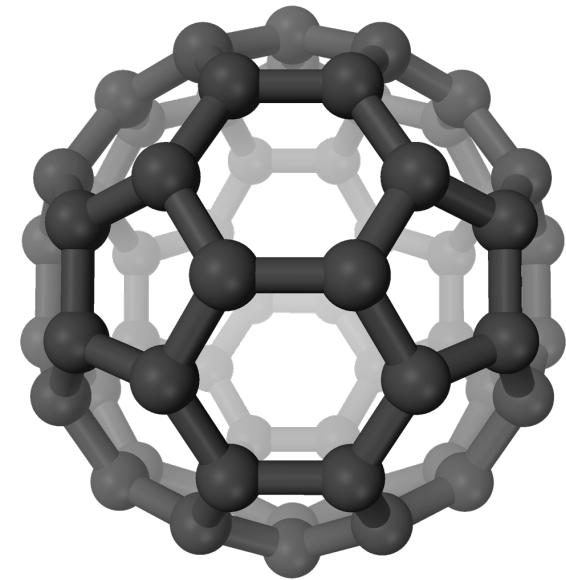
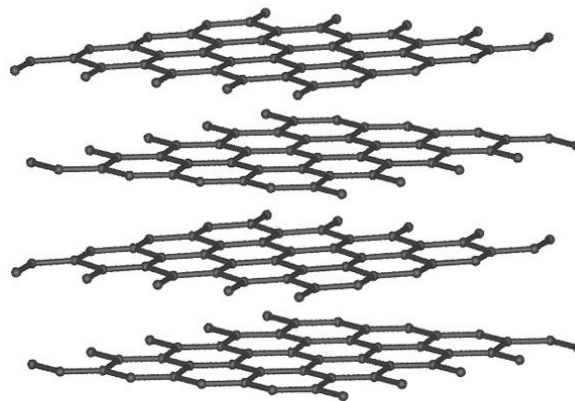
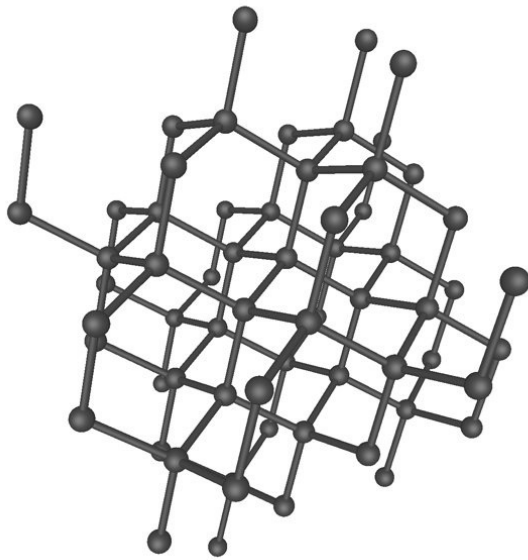
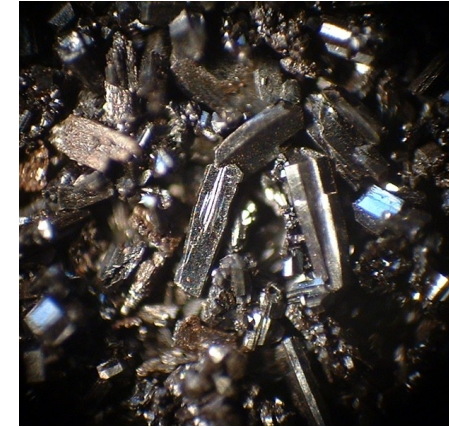
UF₆ and nuclear fuel



Periodic Table

Group ►	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ▼																		Noble gases
Nonmetals	1												Some elements near the dashed staircase are sometimes called <i>metalloids</i>					2
	H												B	C	N	O	F	He
Metals	3	4											5	6	7	8	9	10
	Li	Be											13	14	15	16	17	18
	11	12											Al	Si	P	S	Cl	Ar
	Na	Mg											31	32	33	34	35	36
	19	20											Ga	Ge	As	Se	Br	Kr
	K	Ca											49	50	51	52	53	54
	37	38											In	Sn	Sb	Te	I	Xe
	Rb	Sr											81	82	83	84	85	86
	55	56	La to Yb										81	82	83	84	85	86
	Cs	Ba											Tl	Pb	Bi	Po	At	Rn
	87	88	Ac to No										113	114	115	116	117	118
	Fr	Ra											Nh	Fl	Mc	Lv	Ts	Og
	s-block (incl. He)		f-block	d-block										p-block (excl. He)				
			Lanthanides	57	58	59	60	61	62	63	64	65	66	67	68	69	70	
				La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	
			Actinides	89	90	91	92	93	94	95	96	97	98	99	100	101	102	
				Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	

Allotropes (here of C)



Periodic Table

Group ▶	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18																										
Period ▼																		Noble gases																										
Nonmetals	1												Some elements near the dashed staircase are sometimes called <i>metalloids</i>					2																										
	H												B	C	N	O	F	He																										
Metals	2	3	4											5	6	7	8	9	10																									
	Li	Be											11	12	13	14	15	16	17	18																								
	Na	Mg											19	20	21	22	23	24	25	26	27	28	29	30																				
	K	Ca											31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48														
	Rb	Sr											49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70										
	Cs	Ba	La to Yb											71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101
7	Fr	Ra	Ac to No											103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118															
														Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og															
		s-block (incl. He)		f-block	d-block										p-block (excl. He)																													
				Lanthanides	57	58	59	60	61	62	63	64	65	66	67	68	69	70																										
					La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb																										
				Actinides	89	90	91	92	93	94	95	96	97	98	99	100	101	102																										
					Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No																										

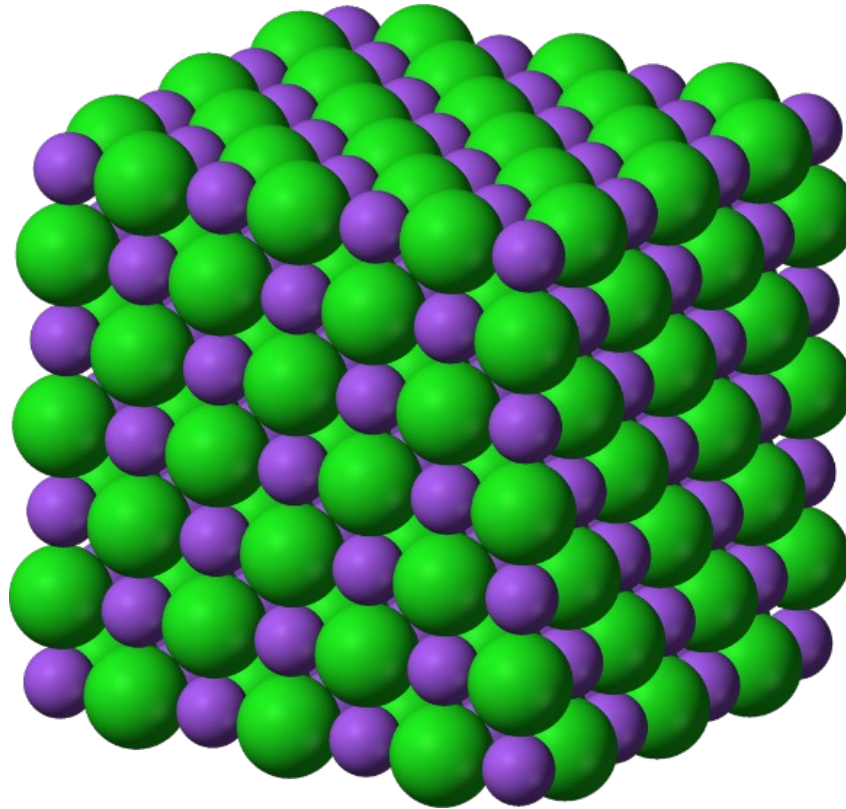
Experiments



Exercise: Balance the reactions.

Structure of ionic compounds

- NaCl is crystalline with Na^+ and Cl^- arranged in a periodic lattice structure
- This enables electrostatic stabilization of the structure



What happens when NaCl get in contact with water?

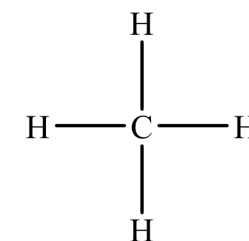
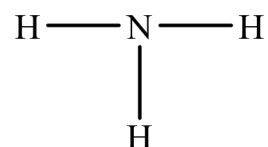
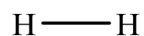
Structure of molecular compounds

Many ways to describe the structure of a molecule:

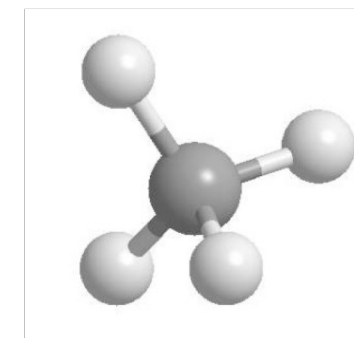
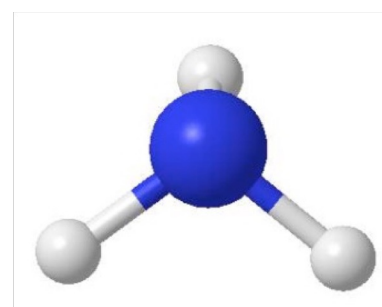
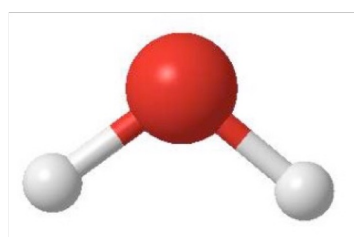
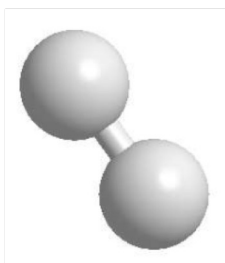
molecular
formula



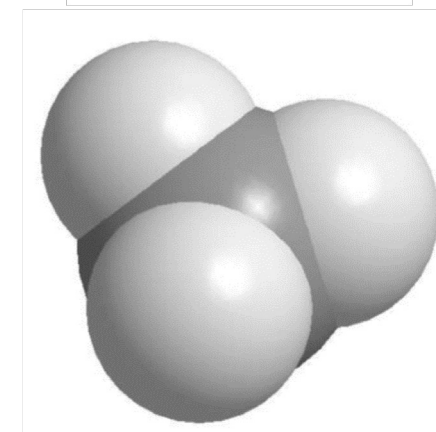
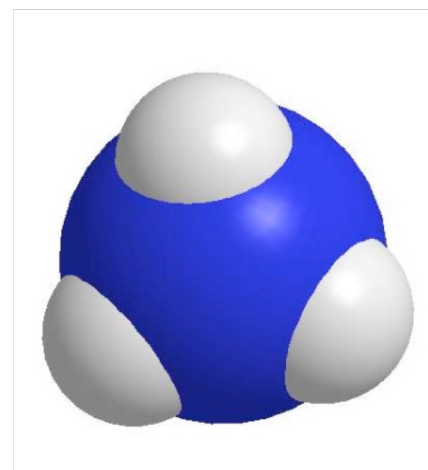
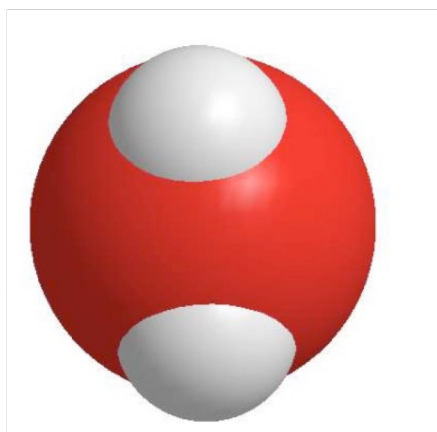
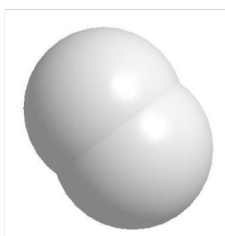
structural
formula



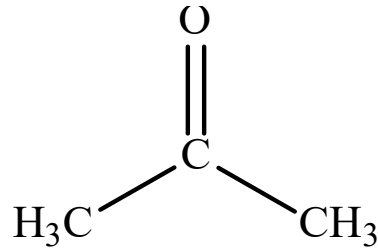
ball-and-
stick
model



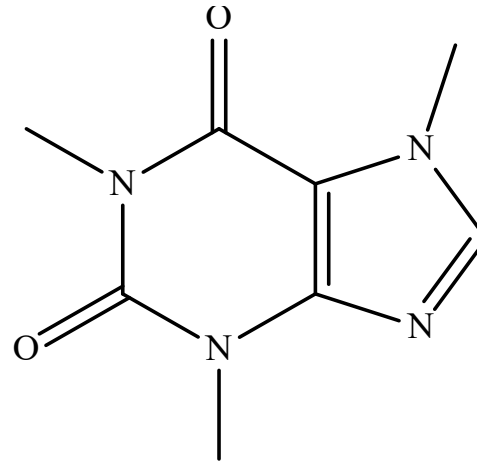
space-
filling
model



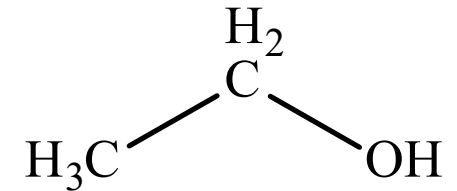
Simplified molecular formulae



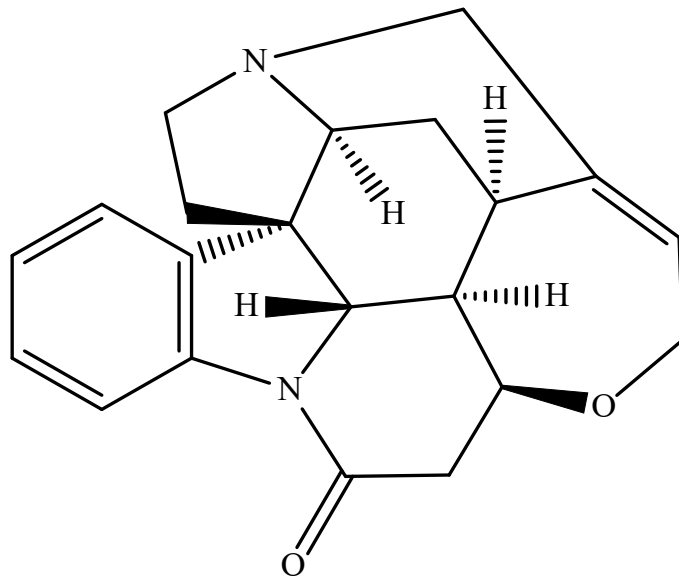
acetone



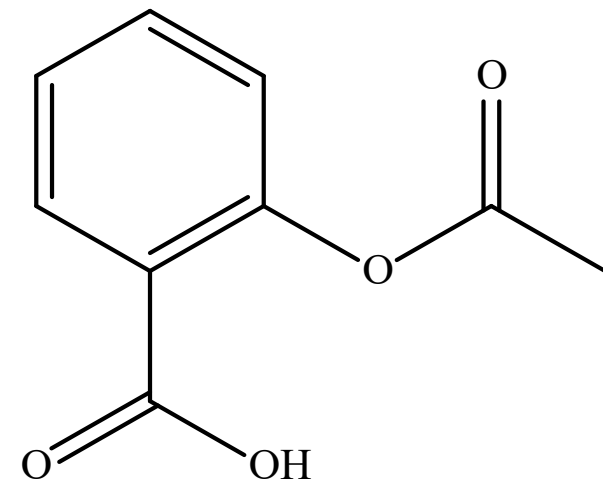
caffeine



ethanol



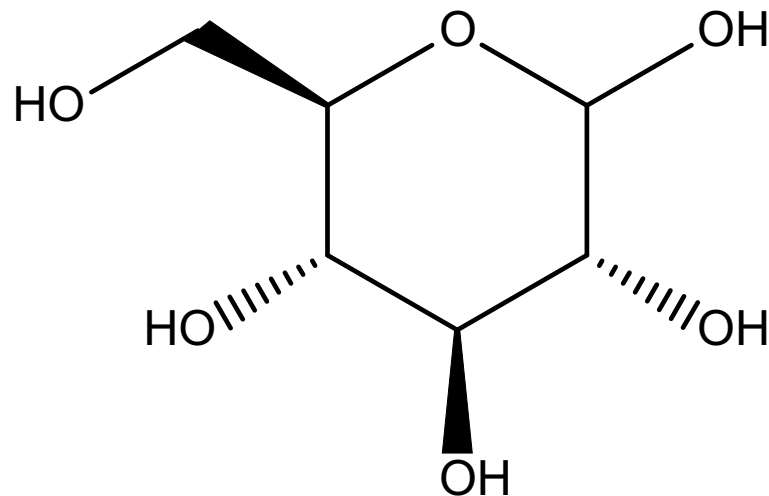
strychnine



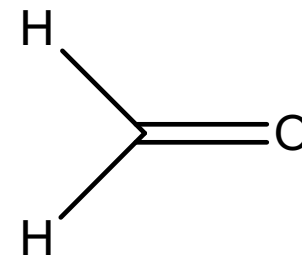
aspirine

Empirical formula

- Only provides the ratio between elements in the molecule (not total count)
- Thus, there will be molecules with the same ratio but different number of atoms in total.
- Both glucose and formaldehyde have the empirical formula, CH_2O :



glucose
 $\text{C}_6\text{H}_{12}\text{O}_6$

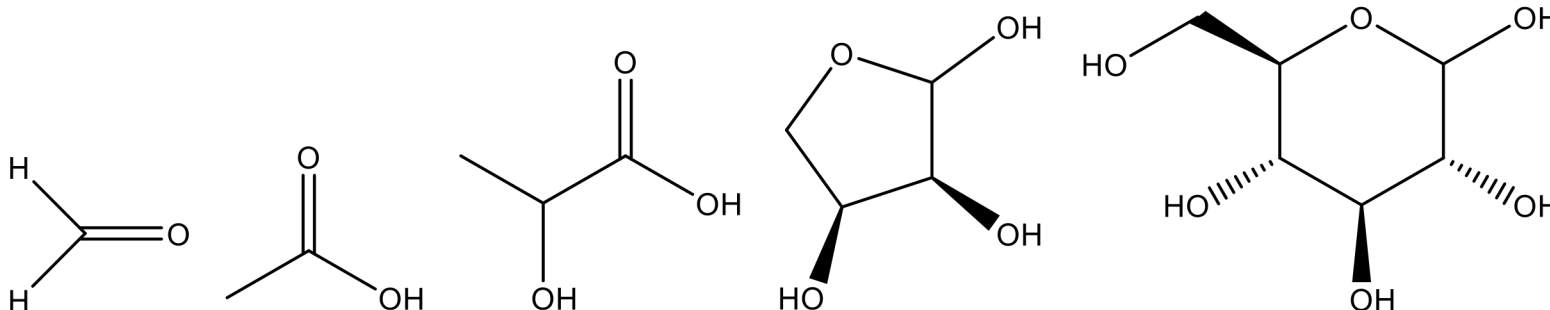


formaldehyde
 CH_2O

More compounds with same empirical formula

Examples of compounds with empirical formula CH_2O (composition by mass 40.0% C, 6.71% H, 53.3% O)

Name	Molecular formula	Whole-number multiple	M/(g/mol)	use or function
Formaldehyde	CH_2O	1	30.03	Disinfectant; biological preservative
Acetic acid	$\text{C}_2\text{H}_4\text{O}_2$	2	60.06	Acetate polymers, vinegar (5% solution); cleaning
lactic acid	$\text{C}_3\text{H}_6\text{O}_3$	3	90.08	causes milk to sour, forms in muscles during anaerobic work; polymers
erythrose	$\text{C}_4\text{H}_8\text{O}_4$	4	120.10	forms during sugar metabolism part of RNA and DNA
ribose	$\text{C}_5\text{H}_{10}\text{O}_5$	5	150.13	part of RNA and DNA and vitamin B ₂
glucose	$\text{C}_6\text{H}_{12}\text{O}_6$	6	180.16	Major sugar in carbohydrate metabolism, the sweetener in some fruits



Period	Hydrogen																Helium		
1	1 H 1.0080																2 He 4.0026		
2	Lithium 3 Li 6.94	Beryl- lium 4 Be 9.0122											Boron 5 B 10.81	Carbon 6 C 12.011	Nitrogen 7 N 14.007	Oxygen 8 O 15.999	Fluorine 9 F 18.998	Neon 10 Ne 20.180	
3	Sodium 11 Na 22.990	Magne- sium 12 Mg 24.305											Alumin- ium 13 Al 26.982	Silicon 14 Si 28.085	Phos- phorus 15 P 30.974	Sulfur 16 S 32.06	Chlorine 17 Cl 35.45	Argon 18 Ar 39.95	
4	Potas- sium 19 K 39.098	Calcium 20 Ca 40.078	Scan- dium 21 Sc 44.956	Tita- nium 22 Ti 47.867	Vana- dium 23 V 50.942	Chrom- ium 24 Cr 51.996	Manga- nese 25 Mn 54.938	Iron 26 Fe 55.845	Cobalt 27 Co 58.933	Nickel 28 Ni 58.693	Copper 29 Cu 63.546	Zinc 30 Zn 65.38	Gallium 31 Ga 69.723	Germa- nium 32 Ge 72.630	Arsenic 33 As 74.922	Sele- nium 34 Se 78.971	Bromine 35 Br 79.904	Krypton 36 Kr 83.798	
5	Rubidium 37 Rb 85.468	Stront- ium 38 Sr 87.62	Yttrium 39 Y 88.906	Zirco- nium 40 Zr 91.224	Niobium 41 Nb 92.906	Molyb- denum 42 Mo 95.95	Tech- netium 43 Tc [97]	Ruthe- nium 44 Ru 101.07	Rho- dium 45 Rh 102.91	Pallad- ium 46 Pd 106.42	Silver 47 Ag 107.87	Cad- mium 48 Cd 112.41	Indium 49 In 114.82	Tin 50 Sn 118.71	Anti- mony 51 Sb 121.76	Tellur- ium 52 Te 127.60	Iodine 53 I 126.90	Xenon 54 Xe 131.29	
6	Caesium 55 Cs 132.91	Barium 56 Ba 137.33	*	Lute- tium 71 Lu 174.97	Haf- nium 72 Hf 178.49	Tanta- lum 73 Ta 180.95	Tung- sten 74 W 183.84	Rhe- nium 75 Re 186.21	Os- mium 76 Os 190.23	Iridium 77 Ir 192.22	Plat- inum 78 Pt 195.08	Gold 79 Au 196.97	Mercury 80 Hg 200.59	Thallium 81 Tl 204.38	Lead 82 Pb 207.2	Bismuth 83 Bi 208.98	Polon- ium 84 Po [209]	Astatine 85 At [210]	Radon 86 Rn [222]
7	Francium 87 Fr [223]	Radium 88 Ra [226]	**	Lawren- cium 103 Lr [266]	Ruther- fordium 104 Rf [267]	Dub- nium 105 Db [268]	Sea- borgium 106 Sg [269]	Bohr- ium 107 Bh [270]	Has- sium 108 Hs [269]	Meit- nerium 109 Mt [278]	Darm- stadtium 110 Ds [281]	Roent- genium 111 Rg [282]	Coper- nicium 112 Cn [285]	Nihon- ium 113 Nh [286]	Flerov- ium 114 Fl [289]	Moscov- ium 115 Mc [290]	Liver- morium 116 Lv [293]	Tenness- ine 117 Ts [294]	Oga- nesson 118 Og [294]
			*	Lan- thanum 57 La 138.91	Cerium 58 Ce 140.12	Praseo- dymium 59 Pr 140.91	Neo- dymium 60 Nd 144.24	Prome- thium 61 Pm [145]	Sama- rium 62 Sm 150.36	Europ- ium 63 Eu 151.96	Gadolin- ium 64 Gd 157.25	Terbium 65 Tb 158.93	Dyspro- sium 66 Dy 162.50	Hol- mium 67 Ho 164.93	Erbium 68 Er 167.26	Thulium 69 Tm 168.93	Ytter- bium 70 Yb 173.05		
			**	Actin- ium 89 Ac [227]	Thorium 90 Th 232.04	Protac- tinium 91 Pa 231.04	Ura- nium 92 U 238.03	Neptu- nium 93 Np [237]	Pluto- nium 94 Pu [244]	Ameri- cium 95 Am [243]	Curium 96 Cm [247]	Berkel- ium 97 Bk [247]	Califor- nium 98 Cf [251]	Einstei- nium 99 Es [252]	Fer- mium 100 Fm [257]	Mende- levium 101 Md [258]	Nobel- ium 102 No [259]		

Definition of “atomic mass”

One atomic mass unit is defined as:
exactly 1/12 of the mass of one ^{12}C atom

Carbon

6

C

12.011

→ Atomic number

→ Element symbol

→ Atomic mass

Why not 12??



Definition of “molar mass”

One *mol* equals the number of atoms in
exactly 12 g of ^{12}C
 $= 6.022 \times 10^{23}$ (Avogadro's number, N_A)

Carbon

6

C

12.011



Atomic number



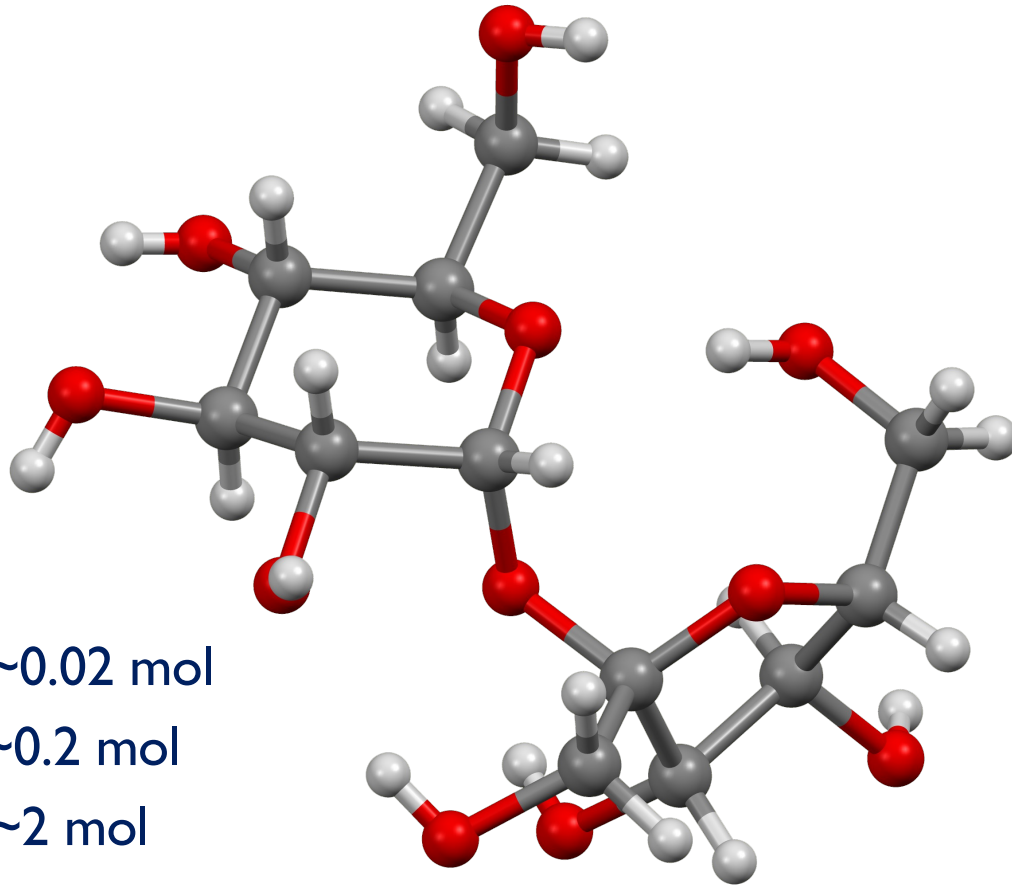
Element symbol



Molar mass
(in g mol^{-1})

What is 1 mol?

How many mol are there approximately in a 1 kg Goldbears? (77% sugar)



- A: ~0.02 mol
- B: ~0.2 mol
- C: ~2 mol
- D: ~2 mol
- E: ~20 mol

Hydrogen	Carbon	Nitrogen	Oxygen
1 H	6 C	7 N	8 O
1.0080	12.011	14.007	15.999

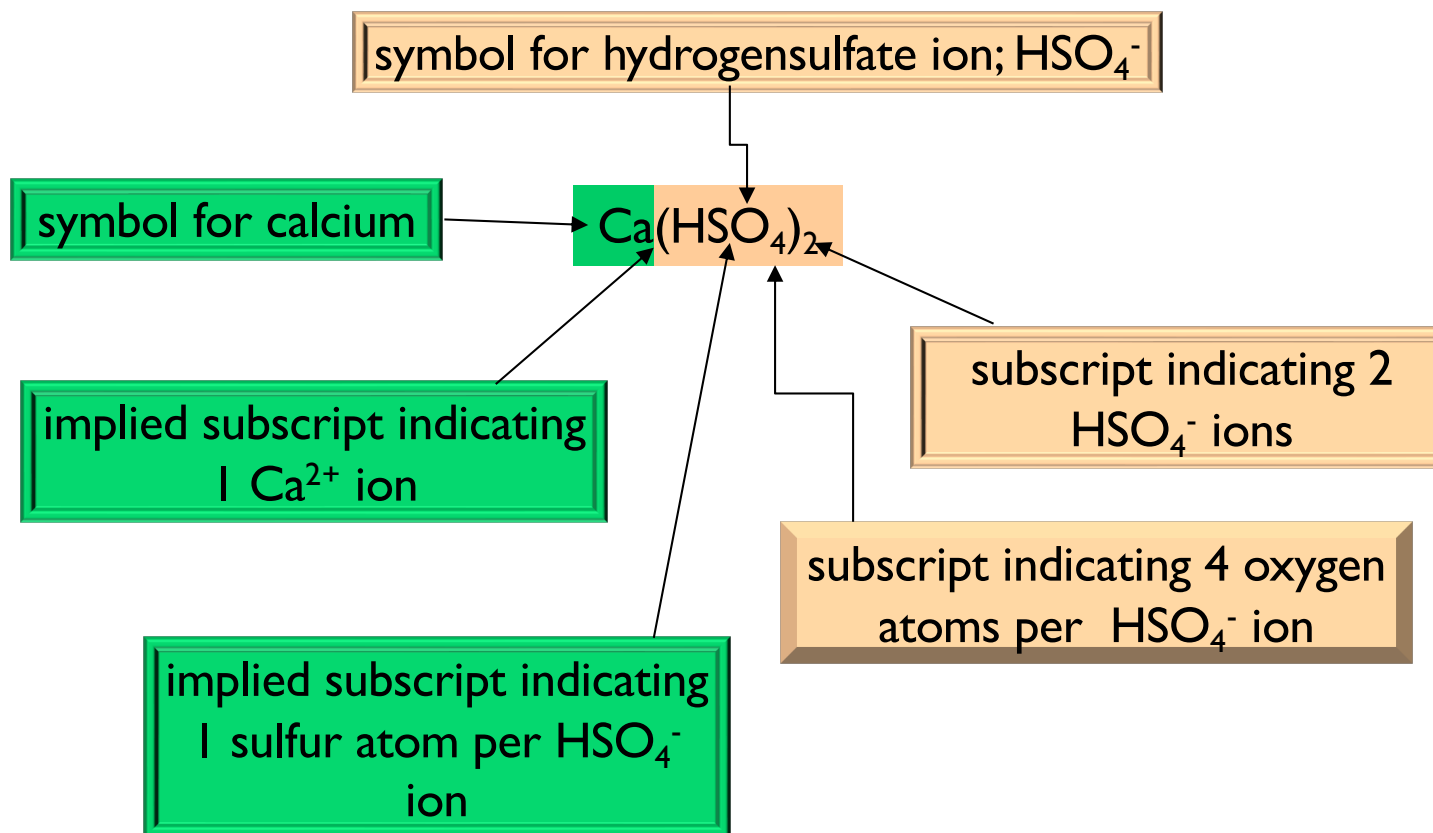


Let's enjoy a snack



- Balance the equation
- How much KClO_3 do I need to combust 1 kg of Goldbears?

Nomenclature



Anions

Some common anions

nonmetal	symbol for ion	base name	anion name
fluorine	F ⁻	fluor-	fluoride
chlorine	Cl ⁻	chlor-	chloride
bromine	Br ⁻	brom-	bromide
iodine	I ⁻	iod-	iodide
oxygen	O ²⁻	ox-	oxide
sulfur	S ²⁻	sulf-	sulfide
nitrogen	N ³⁻	nitr-	nitride

More anions

Some common polyatomic ions

name	formula	name	formula
acetate	CH_3CO_2^-	hypochlorite	ClO^-
carbonate	CO_3^{2-}	chlorate	ClO_2^-
hydrogen carbonate (bicarbonate)	HCO_3^-	chlorite	ClO_3^-
hydroxide	OH^-	perchlorate	ClO_4^-
nitrite	NO_2^-	permanganate	MnO_4^-
nitrate	NO_3^-	sulfite	SO_3^{2-}
chromate	CrO_4^{2-}	hydrogen sulfite (bisulfite)	HSO_3^-
dichromate	$\text{Cr}_2\text{O}_7^{2-}$	sulfate	SO_4^{2-}
phosphate	PO_4^{3-}	hydrogen sulfate (bisulfate)	HSO_4^-
hydrogen phosphate	HPO_4^{2-}	oxide	O^{2-}
ammonium	NH_4^+	cyanide	CN^-

Oxoanions

Some common oxoacids and the corresponding oxoanions

acid	formula	oxyanion	formula
nitrous acid	HNO_2	nitrite	NO_2^-
nitric acid	HNO_3	nitrate	NO_3^-
sulfurous acid	H_2SO_3	sulfite	SO_3^{2-}
sulfuric acid	H_2SO_4	sulfate	SO_4^{2-}
chlorous acid	HClO_2	chlorite	ClO_2^-
chloric acid	HClO_3	chlorate	ClO_3^-
perchloric acid	HClO_4	perchlorate	ClO_4^-
acetic acid	$\text{CH}_3\text{CO}_2\text{H}$	acetate	CH_3CO_2^-
carbonic acid	H_2CO_3	carbonate	CO_3^{2-}

Nomenclature: Molecules

Molecules are usually composed on non-metal elements

In naming, first element of formula is named first, followed by the second one with the addition of “ide”:

HCl Hydrogen chloride

If indices are present, Greek prefixes are used:

CO Carbon monoxide (not ‘monoxide’)

CO₂ Carbon dioxide

N₂O₅ Dinitrogen pentoxide (not ‘pentaoxide’)

SF₆ Sulfur hexafluoride (not ‘hexfluoride’)

mono-	1
di-	2
tri-	3
Tetra-	4
Penta-	5
Hexa-	6
Hepta-	7
Octa-	8
Nona-	9
Deca-	10

Exercise: Name the following



Exercise: Write the formulae

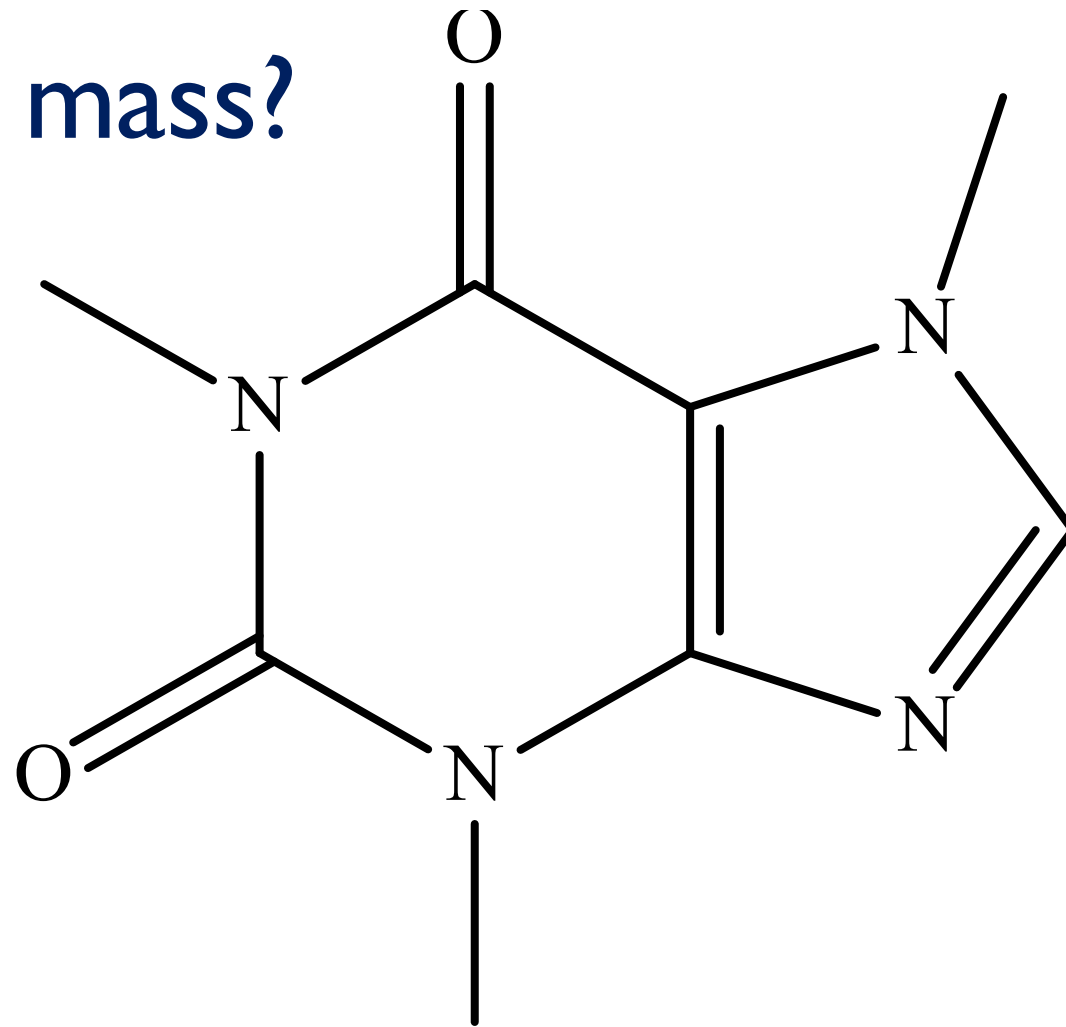
Carbon tetrahydride

Lead(II) bromide chloride

Potassium cyanide

Aluminum sulfate

Test:
Molar mass?



caffeine